A Search for CPT and Lorentz Invariance Violations Using Neutrinos at the NOvA Near Detector

Introduction and Overview

Krishna Lakkaraju, Brian Rebel

Board of Visitors Meeting
22 April 2026

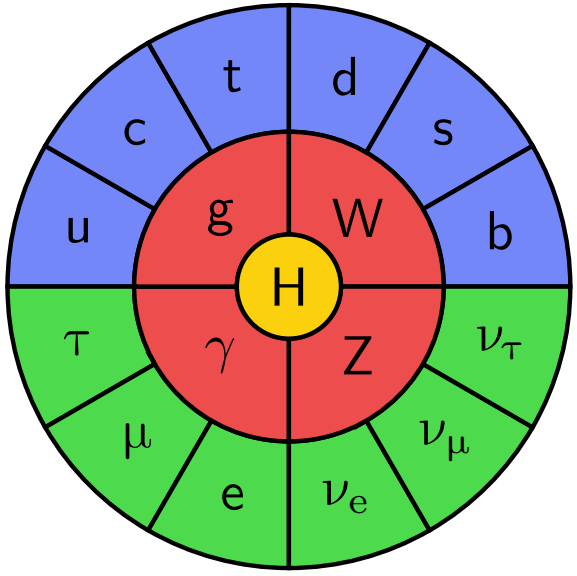


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Neutrinos and the Standard Model

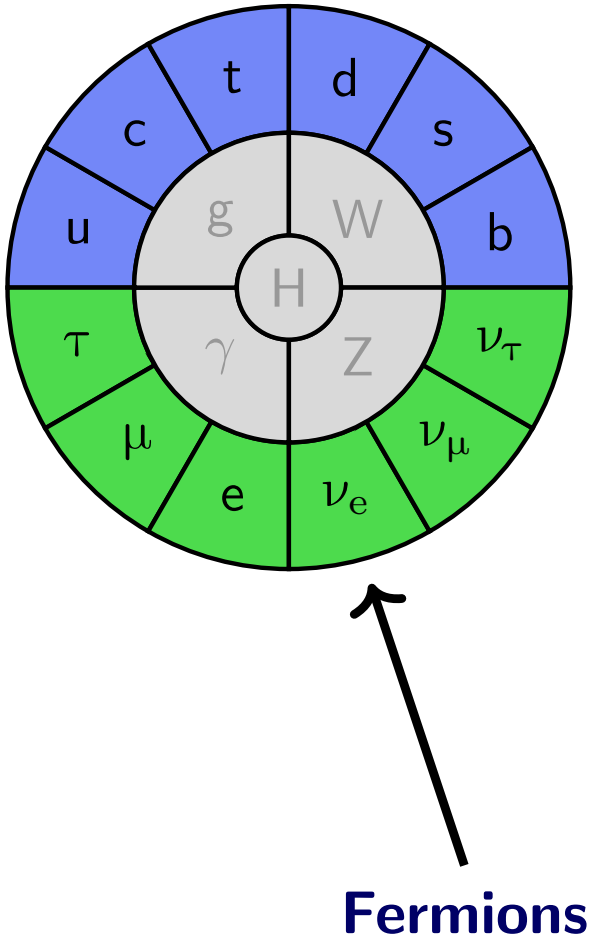
- The Standard Model classifies all known fundamental particles and forces



The Standard Model

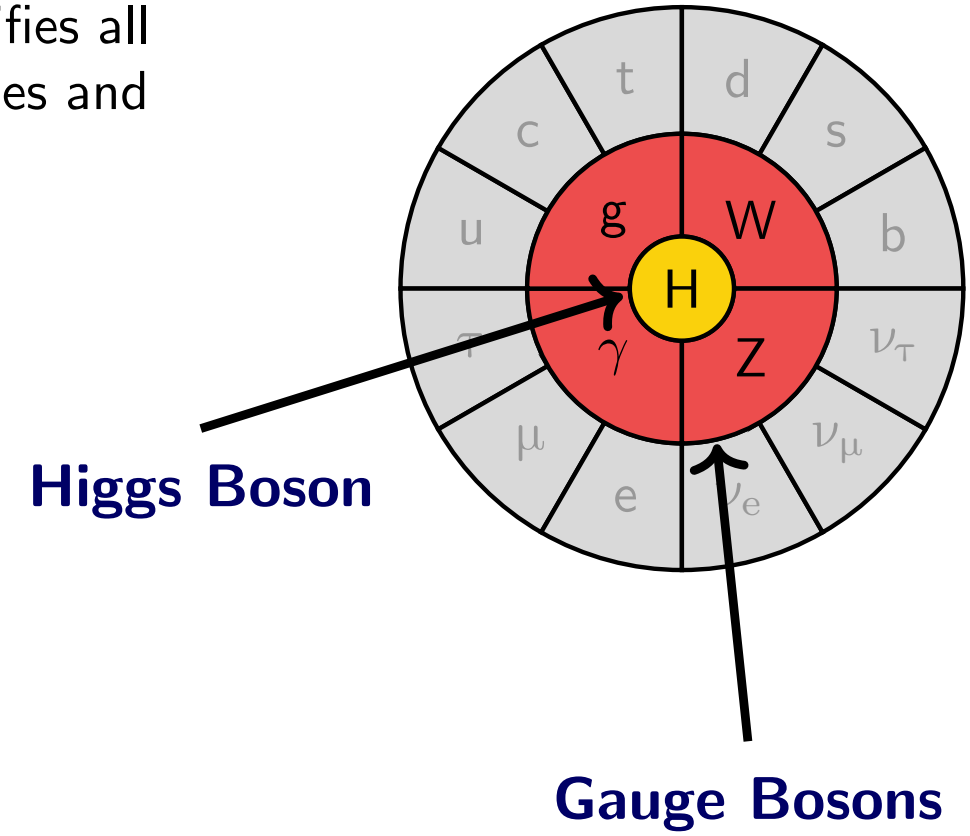
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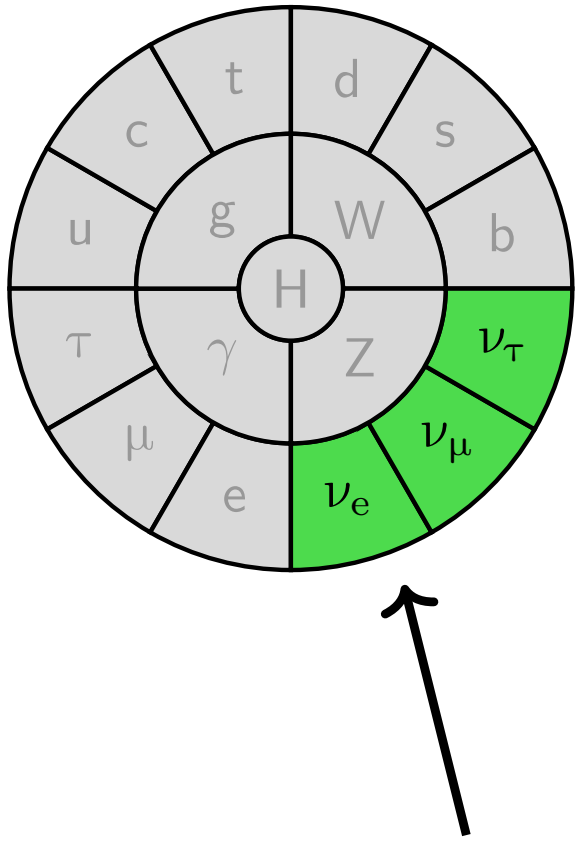
Neutrinos and the Standard Model

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Neutrinos and the Standard Model

- The Standard Model classifies all known fundamental particles and forces
- Neutrinos are the most abundant particle in the universe
- **Neutrinos interact extremely weakly with matter**
- Flavors correspond to the lepton produced in a charged current interaction



Symmetry and the Standard Model

Lorentz Invariance

- **Lorentz Invariance** states that the laws of physics are the same everywhere, at all times
- Lorentz invariance is a core principle of **special relativity**
- Lorentz transforms include coordinate rotations and boosts

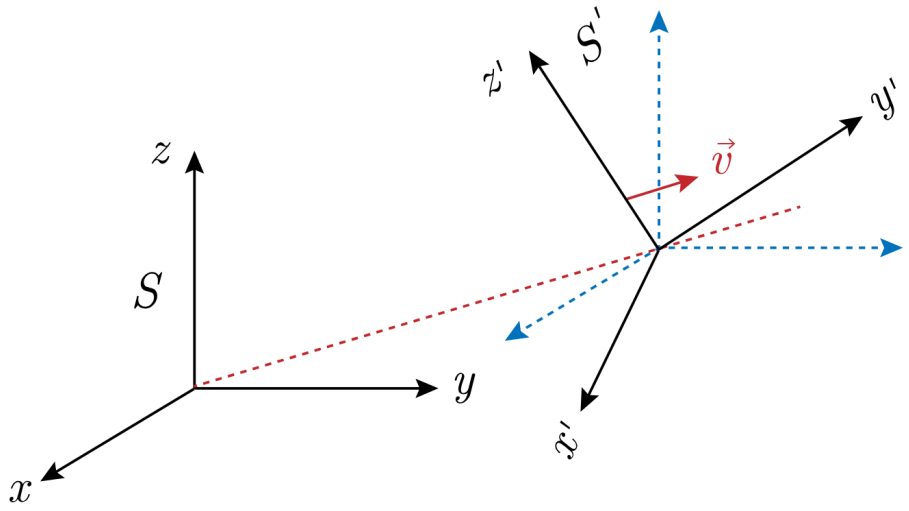


Image adapted from makingphysicsclear.com

CPT Symmetry

- CPT symmetry requires particle interactions to be invariant under the combined operation:
  - **Charge conjugation** (C) inverts the charge of all particles in an interaction
  - **Parity inversion** (P) mirrors the spacial coordinates of an interaction
  - **Time reversal** (T) flips the direction of time evolution in an interaction

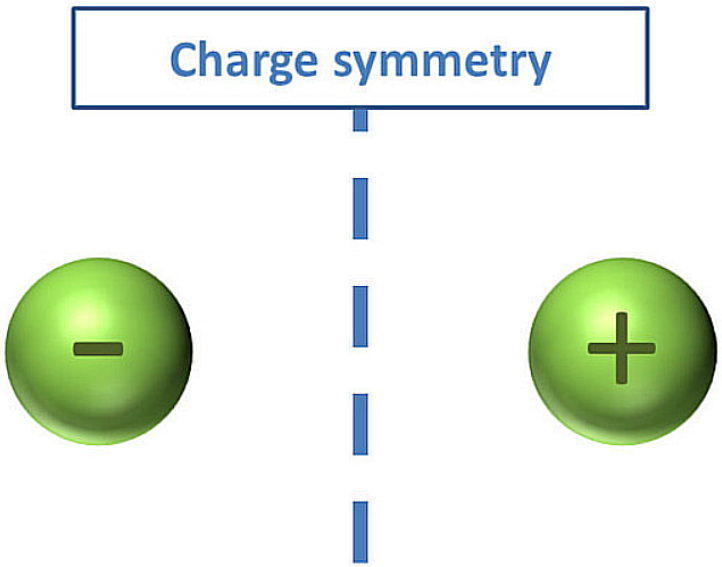


Image adapted from stemit.com (Murphy, 2018)

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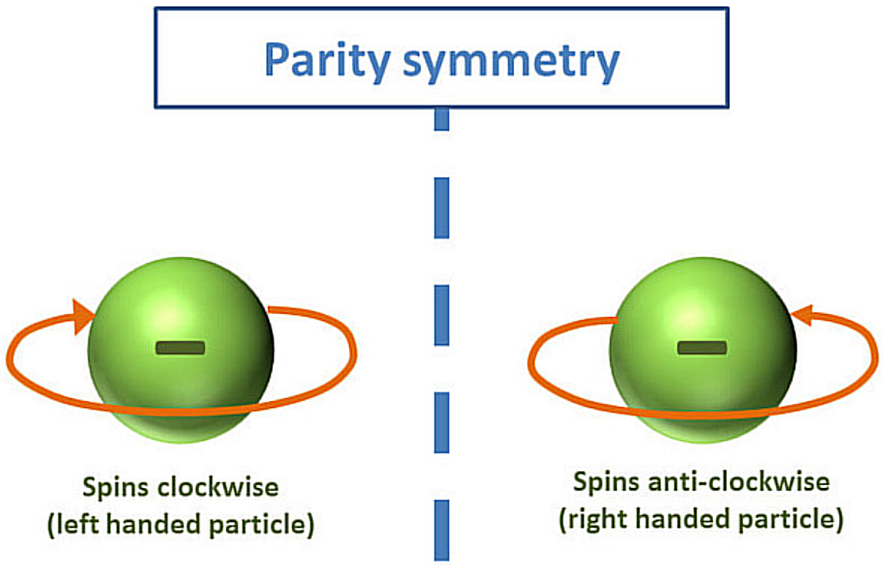


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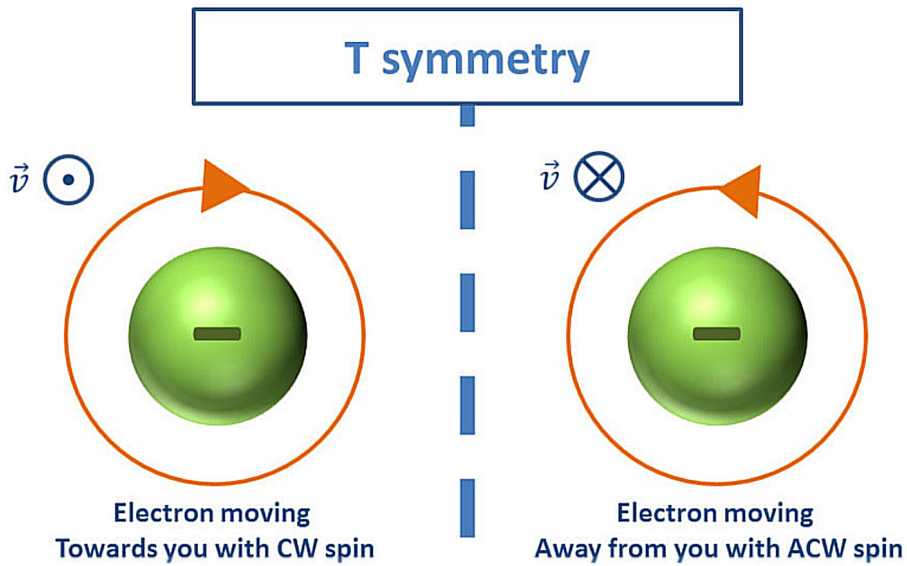


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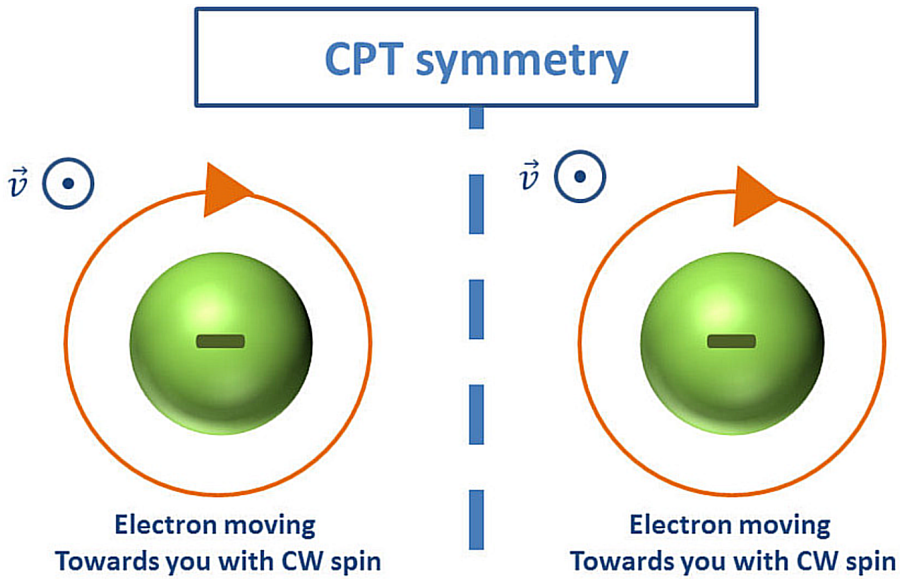


Image adapted from stemit.com (Murphy, 2018)



Since the Standard Model relies on symmetries,

We should test them!

The Standard Model Extension (SME)

What is the SME?

- Includes all possible terms that violate Lorentz and CPT symmetry in an effective Hamiltonian
- Violating terms arise at the Planck scale ($\sim 10^{19}$ GeV).
- Introduces a fixed violating background field

Effective Hamiltonian*

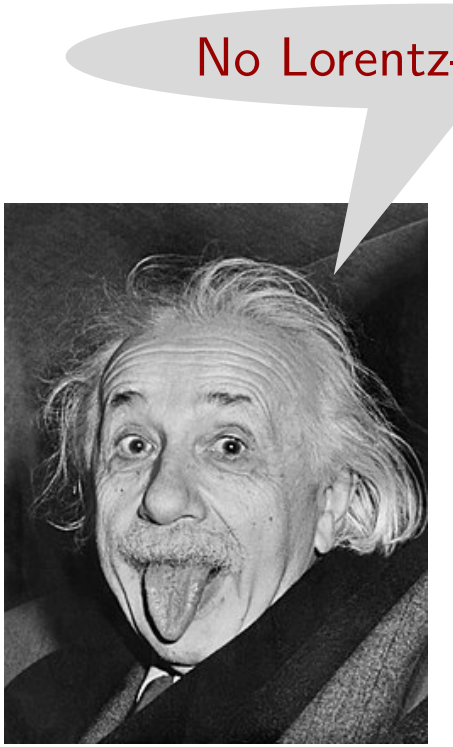
$$\mathcal{H}_{\text{eff}} = \frac{1}{E}\left[(a_L)_\mu p^\mu - (c_L)_{\mu\nu} p^\mu p^\nu\right]$$

- E is the true neutrino energy
- p^μ is the four-momentum of the neutrino
- The magnitude of (a_L) and (c_L) describe the strength of possible Lorentz-violating (LV) effects

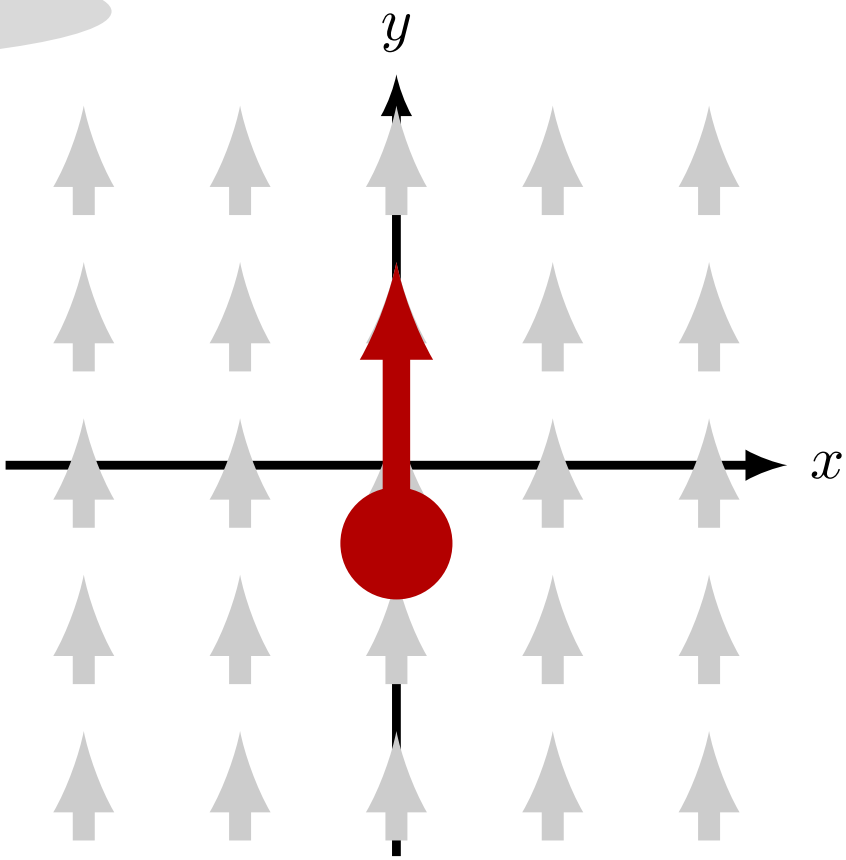
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** Kostelecký and Mewes, Phys. Rev. D* **69**, *016005 (2004), page 2.*

Visualizing Lorentz Invariance Violation



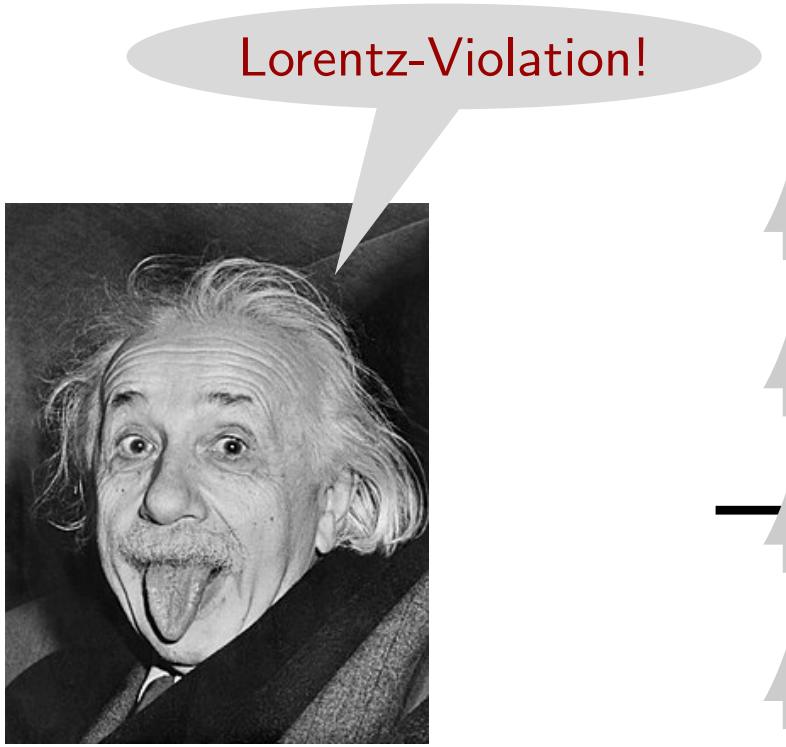
Observer Orientation

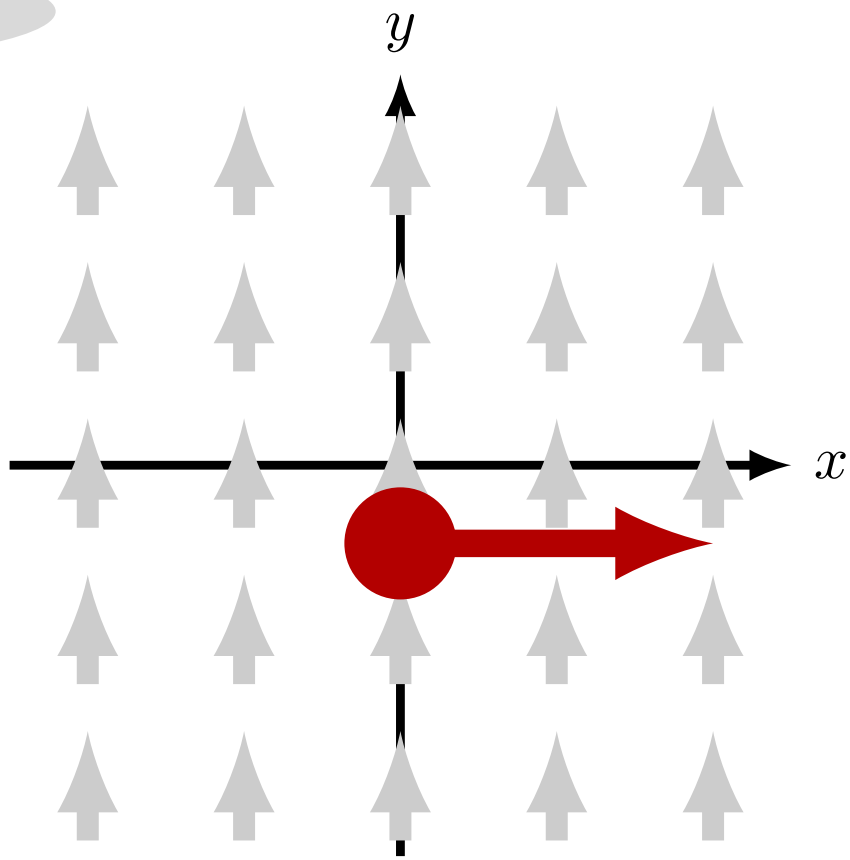


Field-Particle Orientation

Visualization adapted from T. Katori, CPT and Lorentz Symmetry V, World Scientific (2011)

Visualizing Lorentz Invariance Violation



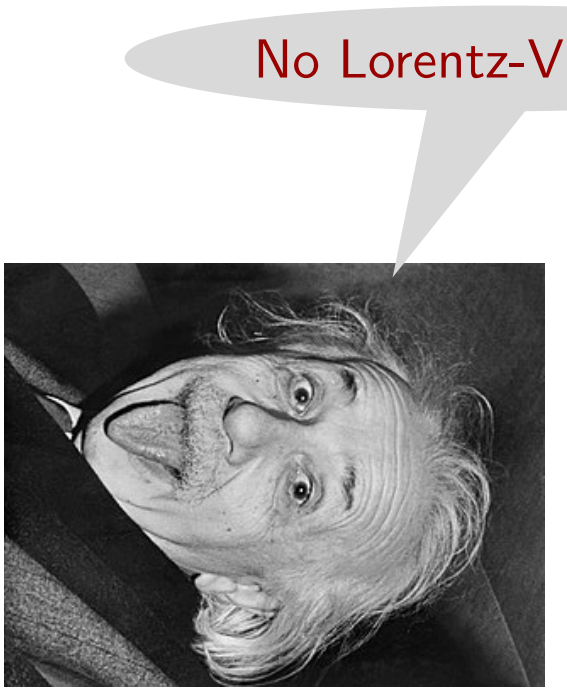


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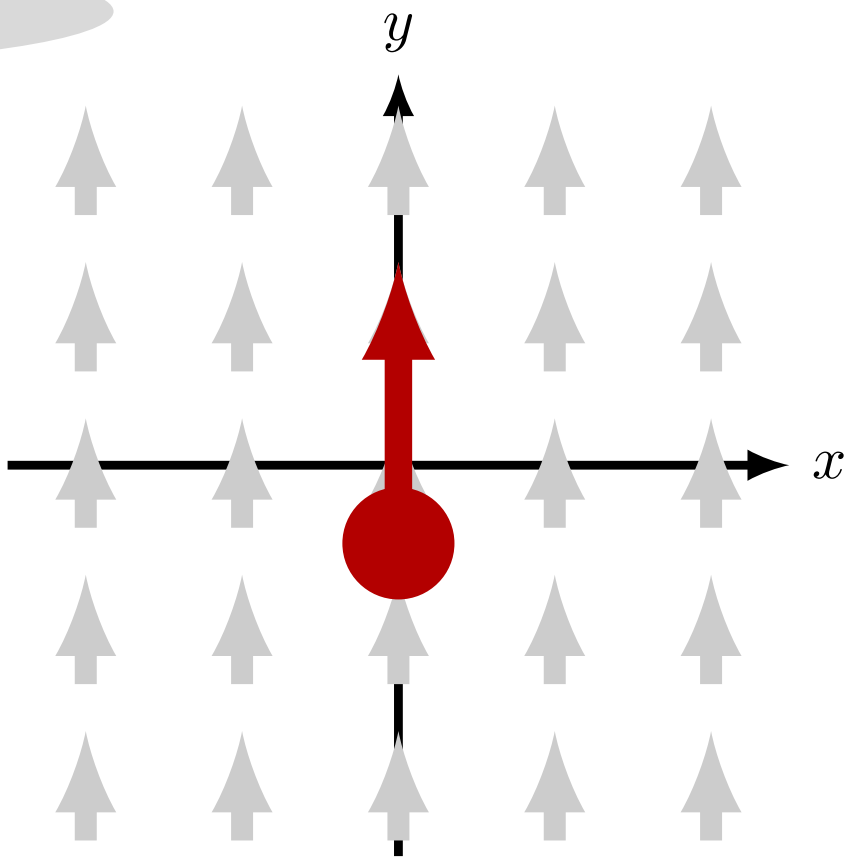
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Visualizing Lorentz Invariance Violation



Observer Orientation



Field-Particle Orientation

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How could we see possible violations?



How could we see possible violations?

Trick: Leverage the Earth's rotation

Lorentz-Violating Background

- The SME characterizes a fixed Lorentz-violating background field
- The entire Earth rotates through the background field
- The violation strength depends on the angle relative to the background
- **LV could cause modulation in any process fixed on the Earth**

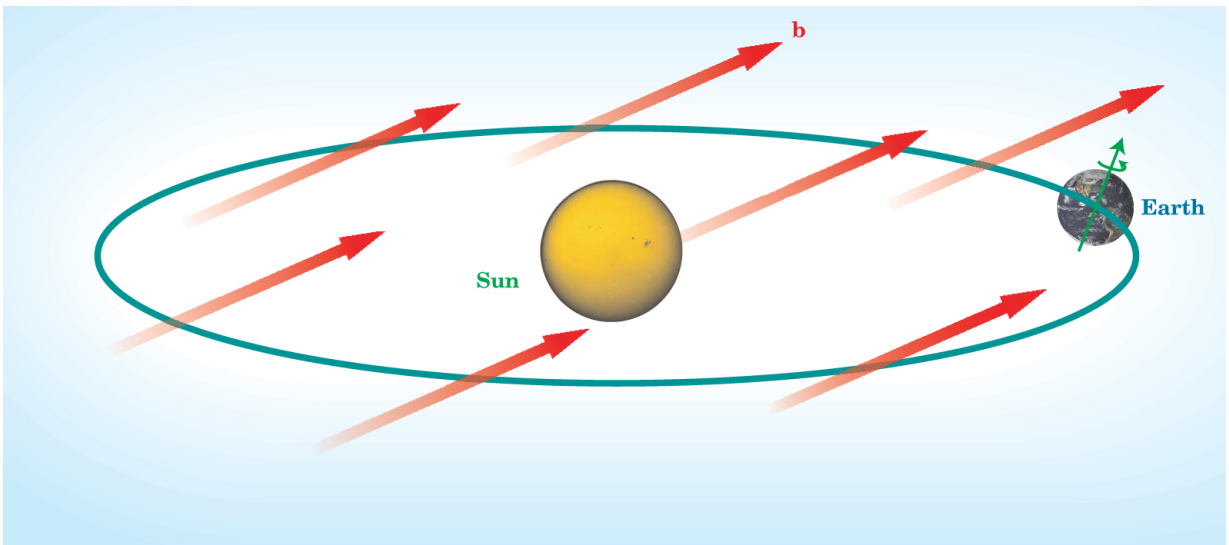
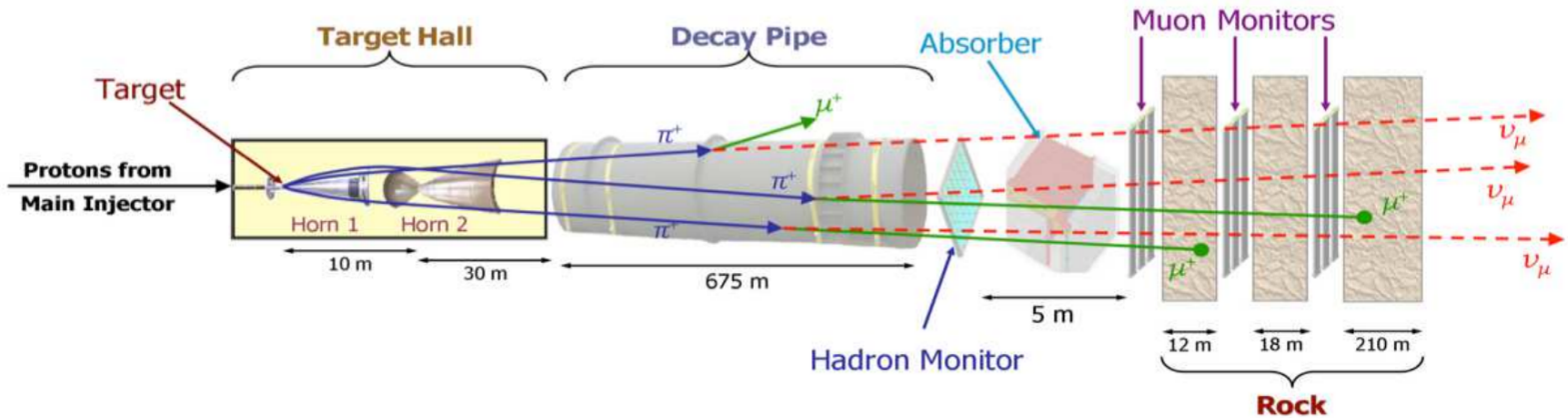


Image adapted from PhysicsToday.org (10.1063/1.1784301)

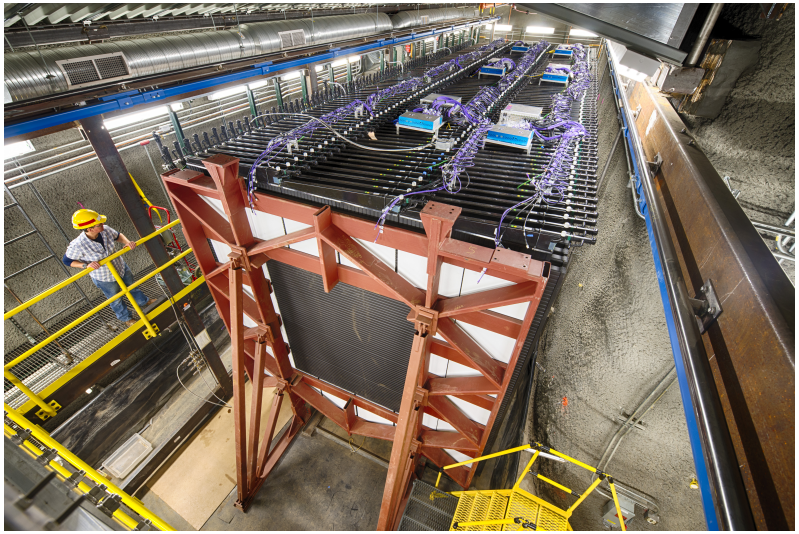
The NOvA Near Detector

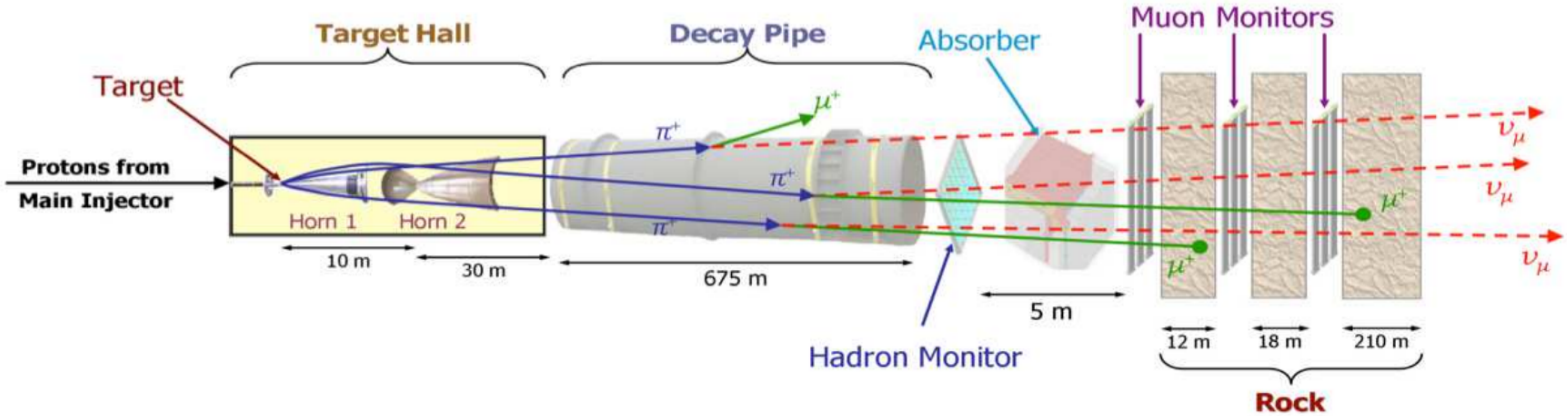
- Utilizes NuMI beam located at FermiLab
- Beam spills deliver protons to the graphite target (POT)
- Beam primarily consists of $\nu_\mu + \bar{\nu}_\mu$ ($\sim 99\%$)



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- Utilizes NuMI beam located at FermiLab
- Beam spills deliver protons to the graphite target (POT)
- Beam primarily consists of $\nu_\mu + \bar{\nu}_\mu$ ($\sim 99\%$)
- The Near Detector sits $\sim$1 km away from the NuMI target





Why NOvA?

- Other neutrino phenomena are negligible at the Near Detector
- Uniquely sensitive to low probability $\nu_\mu \rightarrow \nu_e$ transitions due to the beam's flavor composition
- The large NOvA ν_e sample size may provide improved sensitivity to Lorentz-violating SME coefficients
- This study is **the first search** for Lorentz-Invariance violations at NOvA

$\nu_\mu \rightarrow \nu_{e,\tau}$

Previously Studied[1]

$\nu_\mu \rightarrow \nu_e \qquad \nu_e \rightarrow \nu_{\mu,\tau}$

Channels for Possible Improvement[2]

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[1] *Adamson et al., Phys. Rev. Lett.* **105**, *151601 (2010)*

[2] *A. A. Aguilar-Arevalo et al., Phys. Lett. B* **718**, *1303–1308 (2013)*

Sidereal Time

- A **solar day** is the time it takes for the Sun to return to the same position in the sky
- A **sidereal day** is the length of one rotation about the Earth's axis
- The sidereal day is approximately four minutes shorter than the solar day

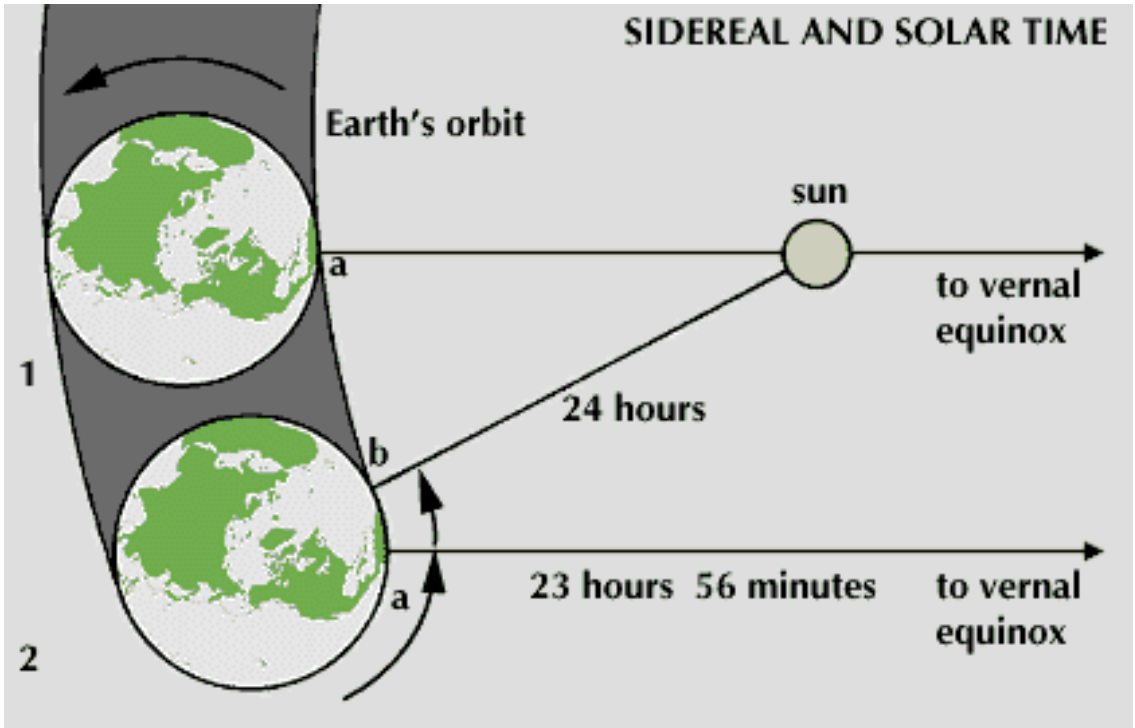


Image adapted from www.schoolphysics.co.uk (Solar and Sidereal Days).

Sidereal Modulation

- Rotation of the NuMI beam in the background field allows access to possible Lorentz-violating effects
- **If LV exists, we will observe these effects as sidereal modulations on the neutrino event rate**

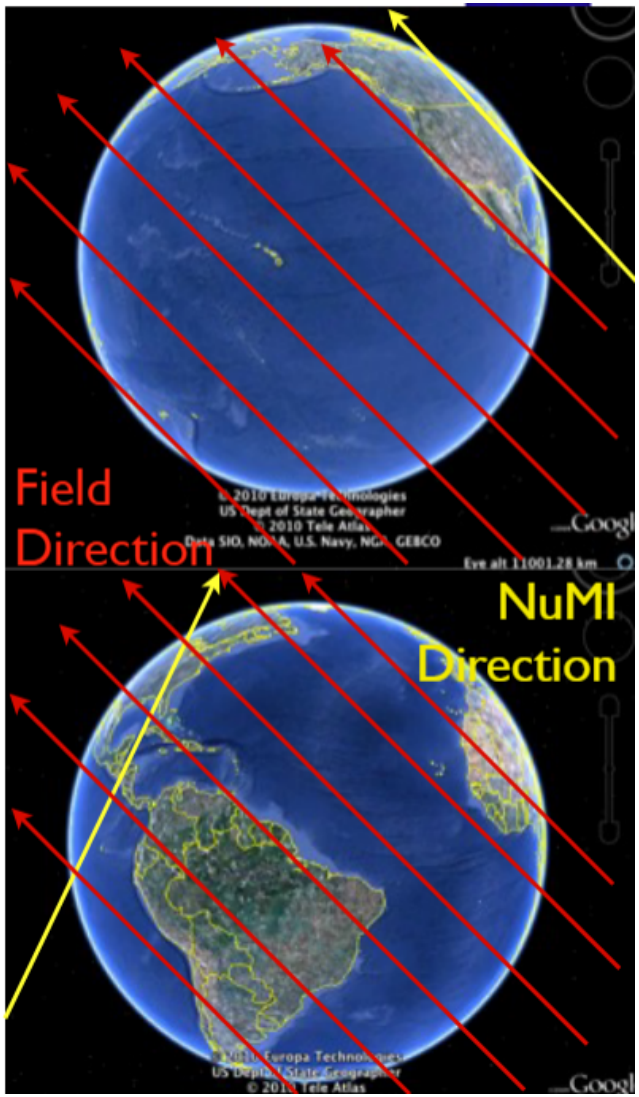
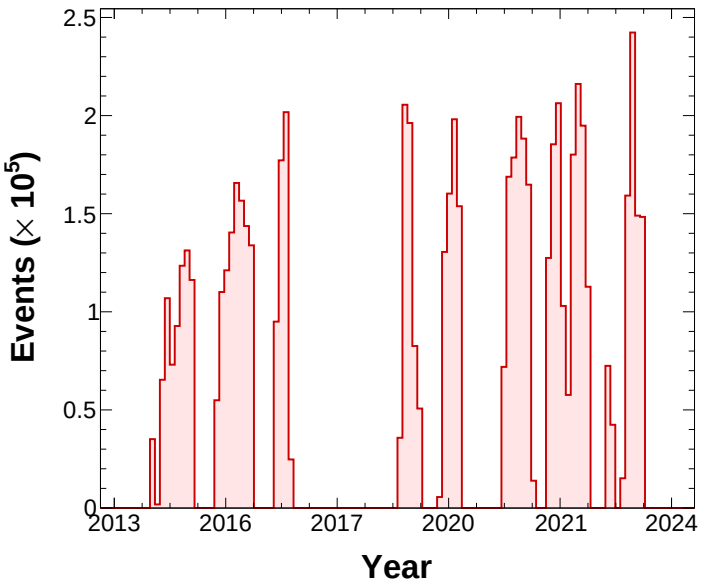


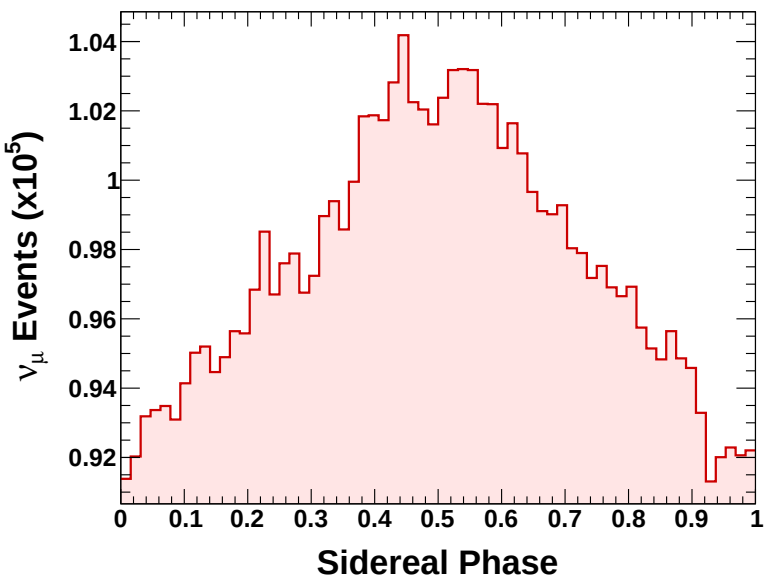
Image courtesy of Prof. Brian Rebel, UW–Madison ν-group, 2011.

Events in the Detector

- We use selected neutrinos at the Near Detector between 2014-2022
- Events are initially tagged with a UNIX time, converted to sidereal phase
- Selected events only occur during beam spills
- The number of neutrinos we observe is proportional to POT

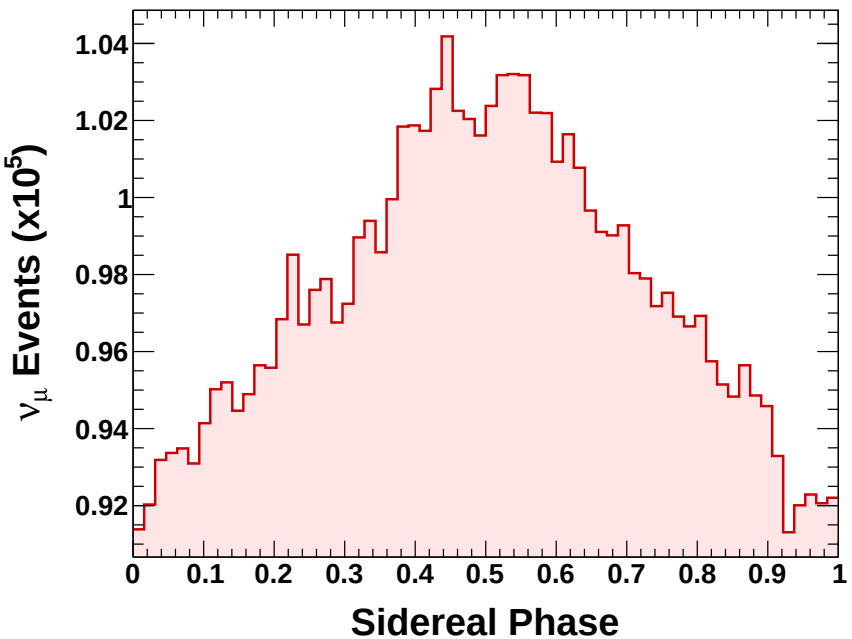


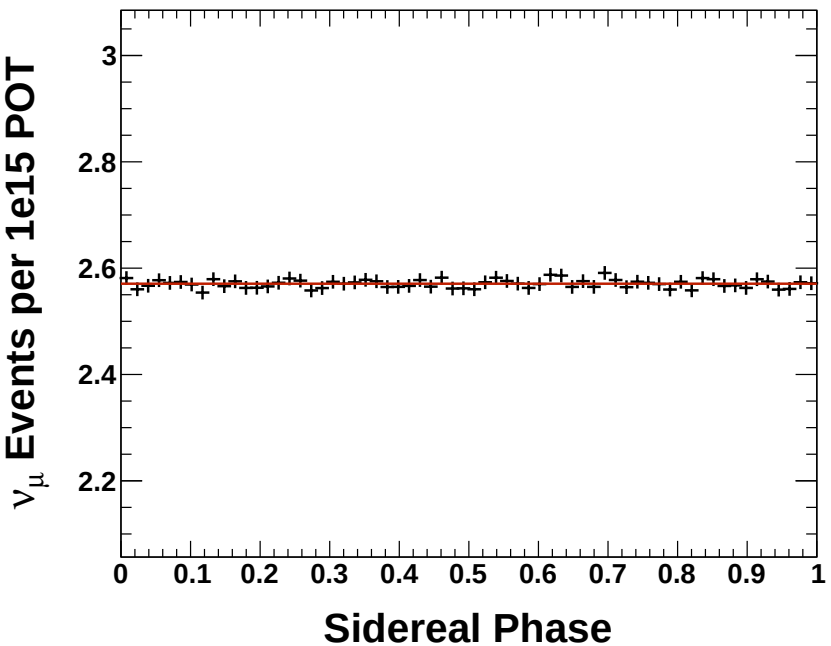
Convert to LSP



Neutrino Event Rate

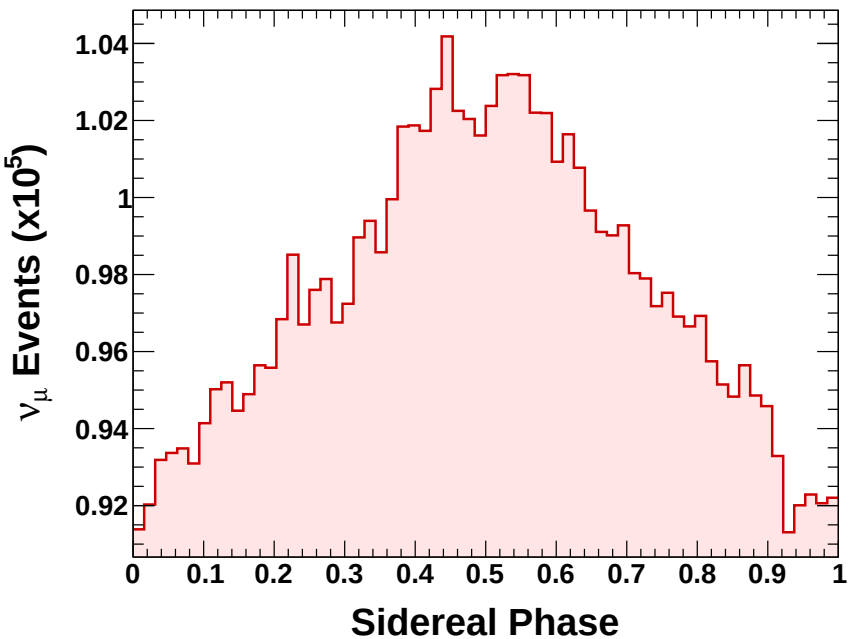
- The event rate is the number of events per POT during some time interval
- We can measure the amount of modulation in the event rate

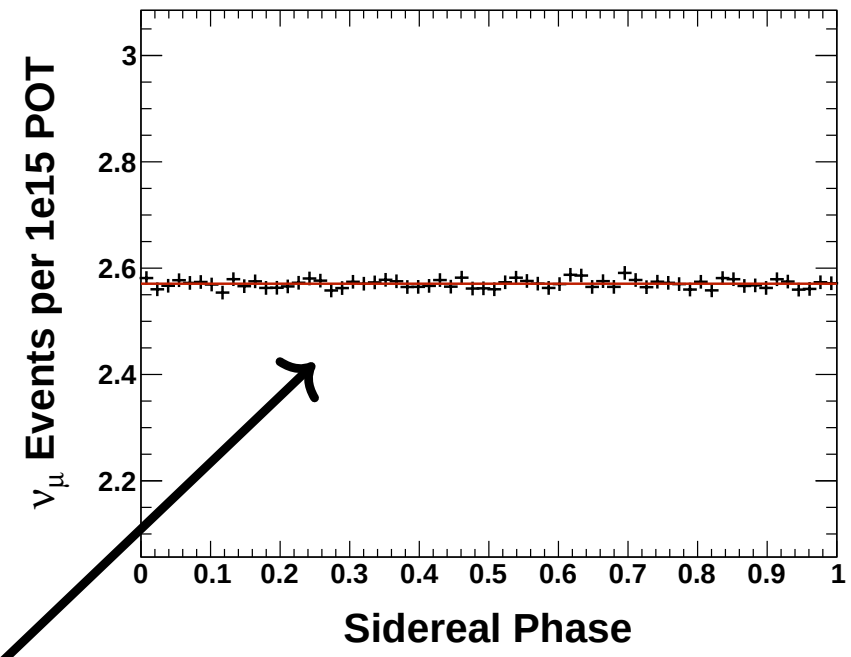




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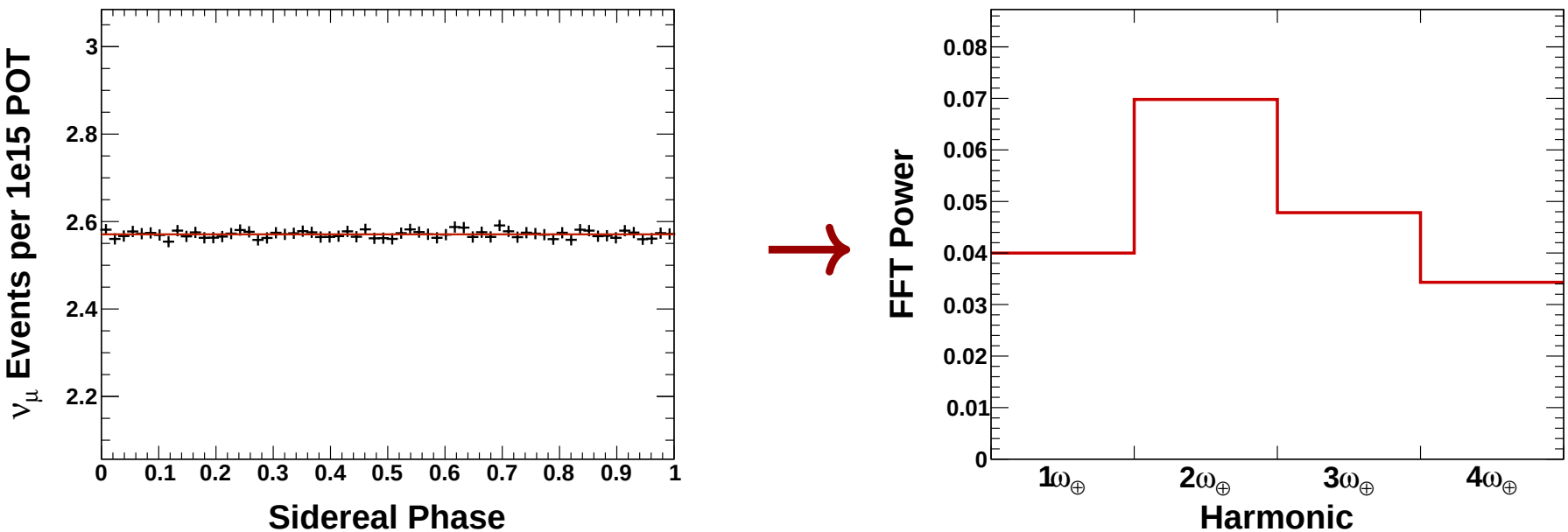




No LV Signal!

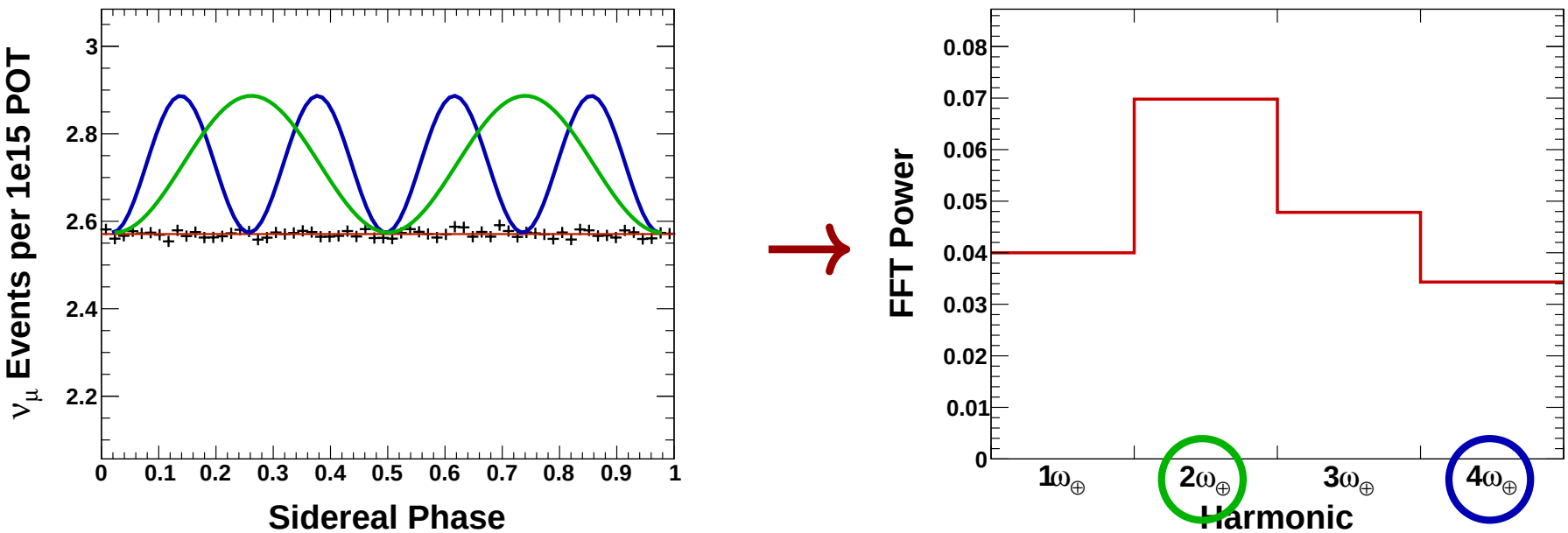
Fast Fourier Transform

- The Fast Fourier Transform (FFT) converts signals in time to powers in the frequency domain
- FFT power measures the signal strength for each harmonic



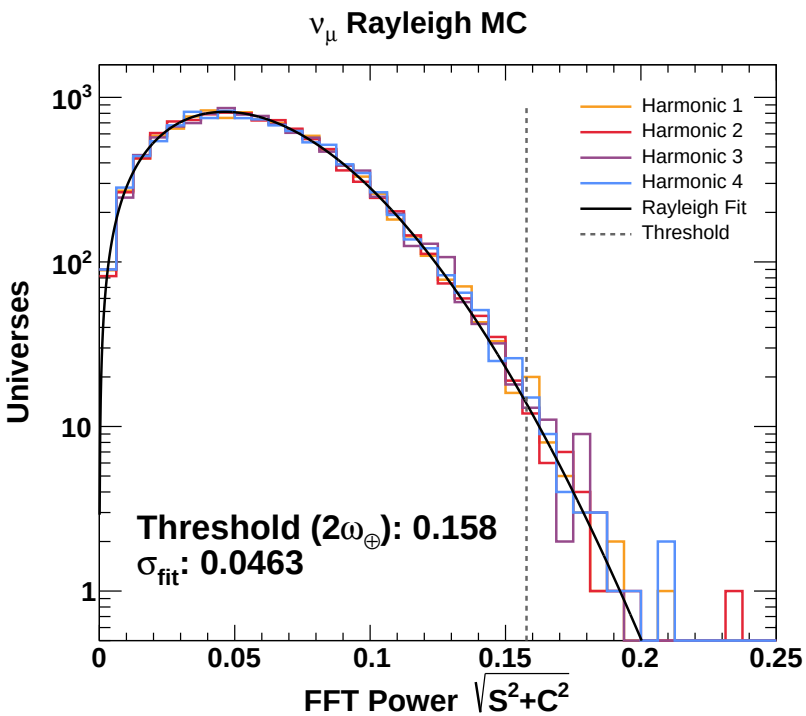
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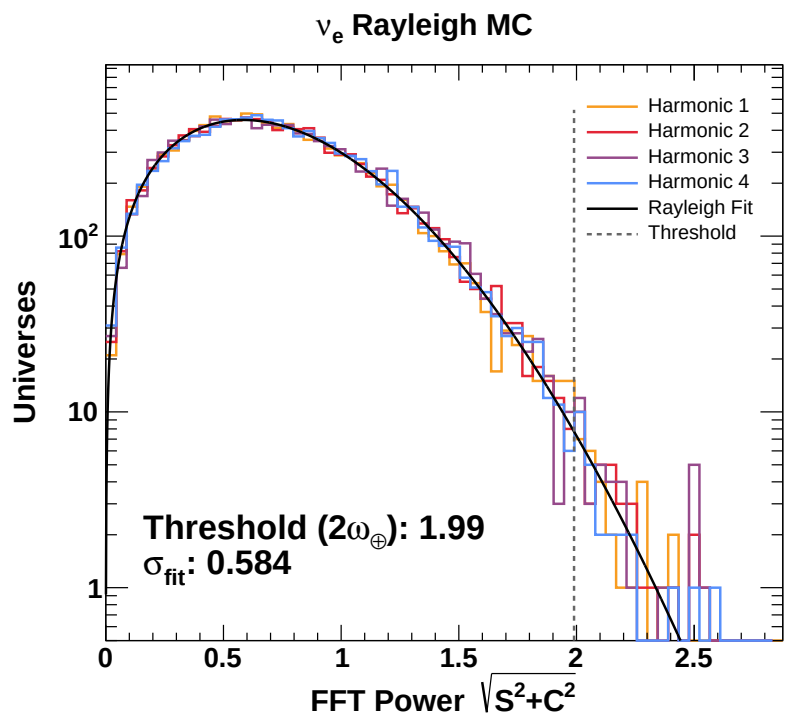
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Signal Threshold

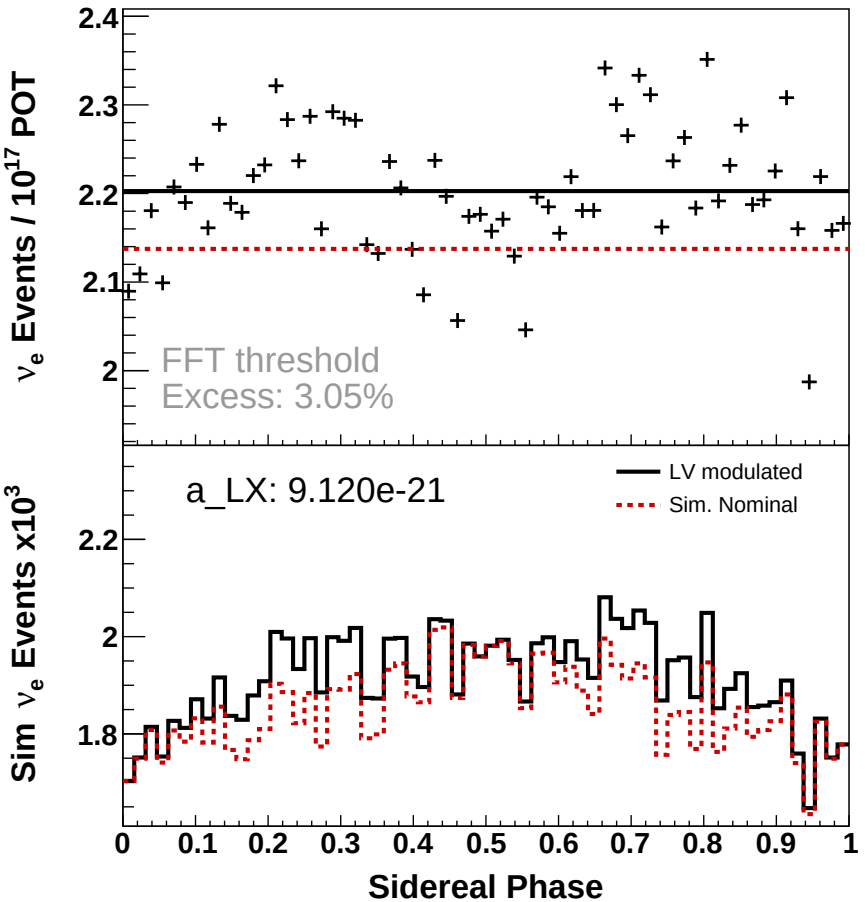
- A signal threshold quantifies the size of FFT powers
- We remove any sidereal modulation by randomizing spill timestamps
- Build a distribution of possible power values by simulating many pseudo-experiments
- We record the first four harmonics in each pseudo-experiment





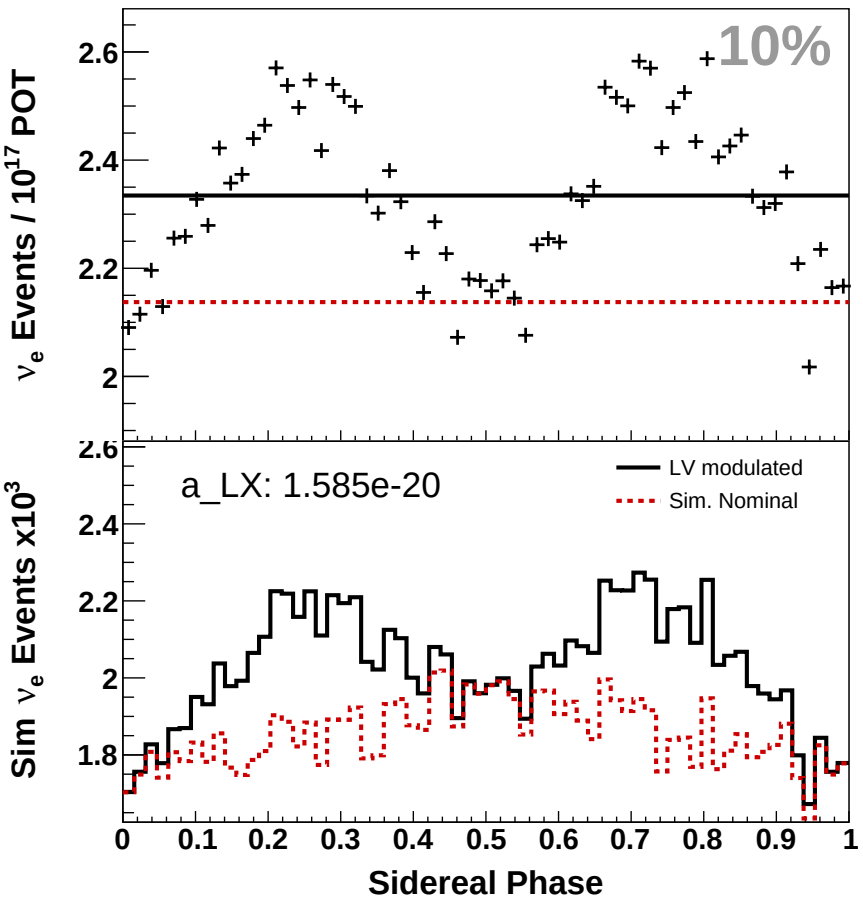
Establishing SME Limits

- Set one SME coefficient non-zero to inject LV signal
- For each event, compute LV-driven oscillation weight, $P_{\mathrm{LV}}(E, L, \omega_{\oplus} T_{\oplus})$
- Determine the smallest coefficient value where FFT signal $>$ threshold



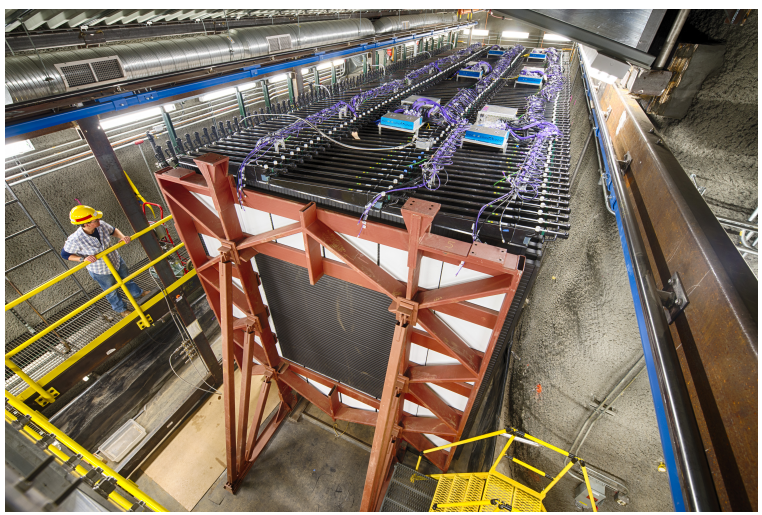
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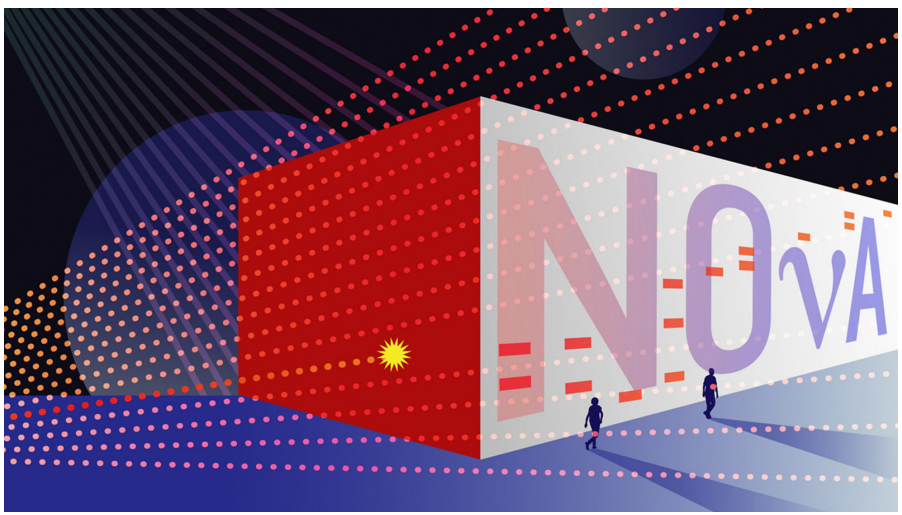
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- Figure shows what LV would look like for 10% excess ν_e events.



Summary

- Lorentz invariance and CPT symmetry are fundamental to the Standard Model
- Any observed violations would introduce new physics
- The NOvA experiment is ideal for searching for tiny violations
- UW-Madison is leading a search at the NOvA Near Detector





Thank you for listening!

We gratefully acknowledge the many contributions and support from the members of the UW-Madison ν-group, NOvA Collaboration, FermiLab staff, and US DoE.



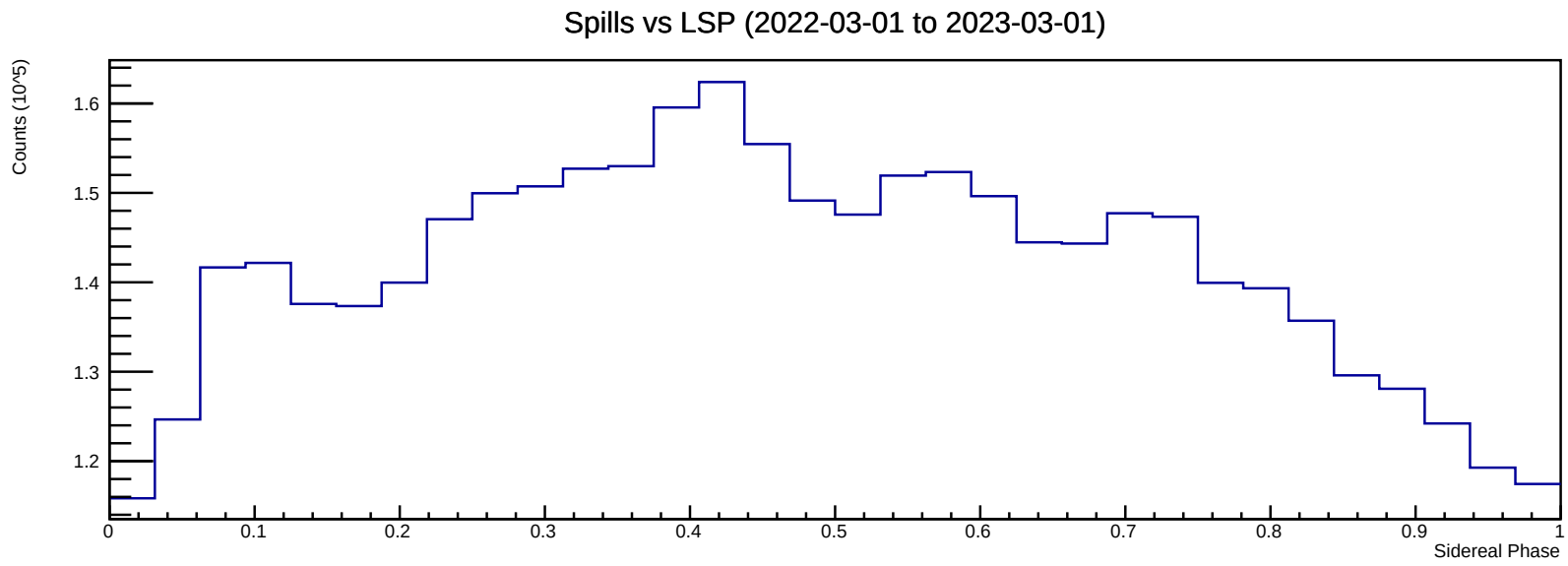


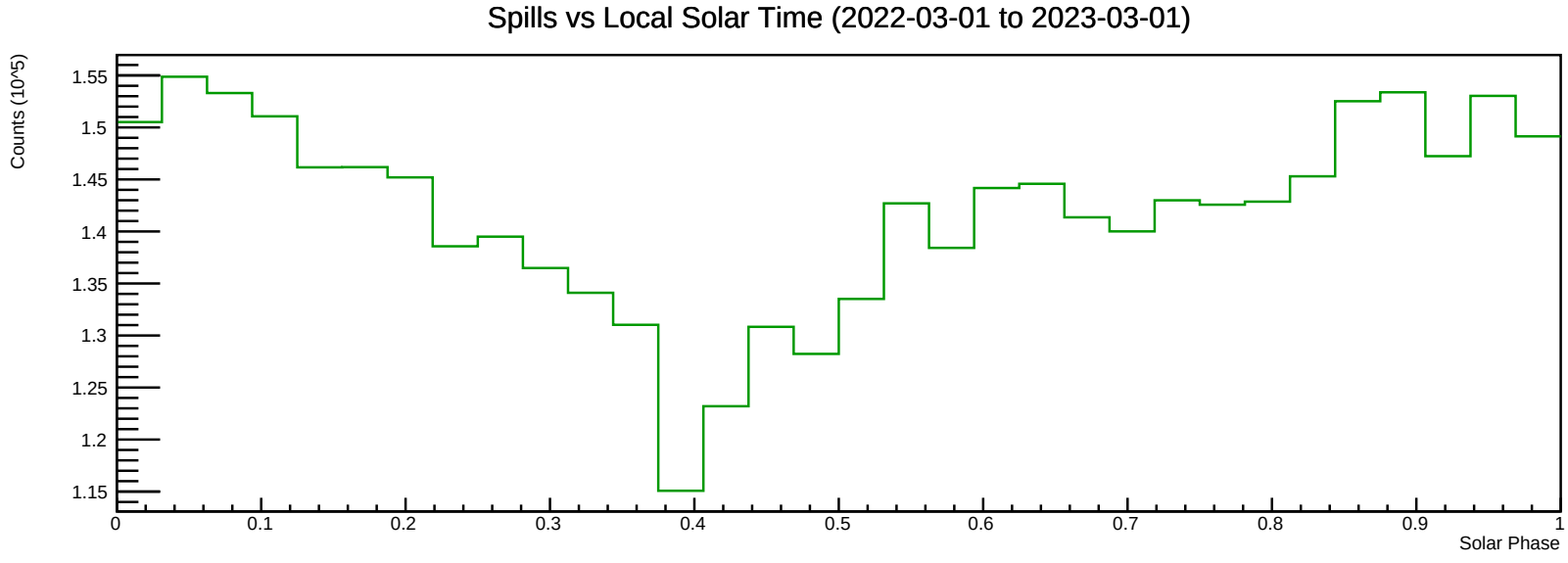
Backup: SME Intermediate Calculations

$$
\begin{aligned}
(\mathcal{A}_s^{(0)})_{ab} &= \hat{N}^Y (a_L)^X_{ab} - \hat{N}^X (a_L)^Y_{ab}, \\
(\mathcal{A}_s^{(1)})_{ab} &= -2\hat{N}^Y (c_L)^{TX}_{ab} + 2\hat{N}^X (c_L)^{TY}_{ab} \\
&\quad + 2\hat{N}^Y \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^X \hat{N}^Z (c_L)^{YZ}_{ab}, \\
(\mathcal{A}_c^{(0)})_{ab} &= -\hat{N}^X (a_L)^X_{ab} - \hat{N}^Y (a_L)^Y_{ab}, \\
(\mathcal{A}_c^{(1)})_{ab} &= 2\hat{N}^X (c_L)^{TX}_{ab} + 2\hat{N}^Y (c_L)^{TY}_{ab} \\
&\quad - 2\hat{N}^X \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^Y \hat{N}^Z (c_L)^{YZ}_{ab}, \\
(\mathcal{B}_s^{(1)})_{ab} &= \hat{N}^X \hat{N}^Y \left((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\right) \\
&\quad - \left(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\right)(c_L)^{XY}_{ab}, \\
(\mathcal{B}_c^{(1)})_{ab} &= -\tfrac{1}{2}\left(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\right)\left((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\right) \\
&\quad - 2\hat{N}^X \hat{N}^Y (c_L)^{XY}_{ab}.
\end{aligned}
$$

$$
\begin{aligned}
(\mathcal{C})_{ab} &= (\mathcal{C}^{(0)})_{ab} + E(\mathcal{C}^{(1)})_{ab}, \\
(\mathcal{A}_s)_{ab} &= (\mathcal{A}_s^{(0)})_{ab} + E(\mathcal{A}_s^{(1)})_{ab}, \\
(\mathcal{A}_c)_{ab} &= (\mathcal{A}_c^{(0)})_{ab} + E(\mathcal{A}_c^{(1)})_{ab}, \\
(\mathcal{B}_s)_{ab} &= E(\mathcal{B}_s^{(1)})_{ab}, \quad (\mathcal{B}_c)_{ab} = E(\mathcal{B}_c^{(1)})_{ab}
\end{aligned}
$$

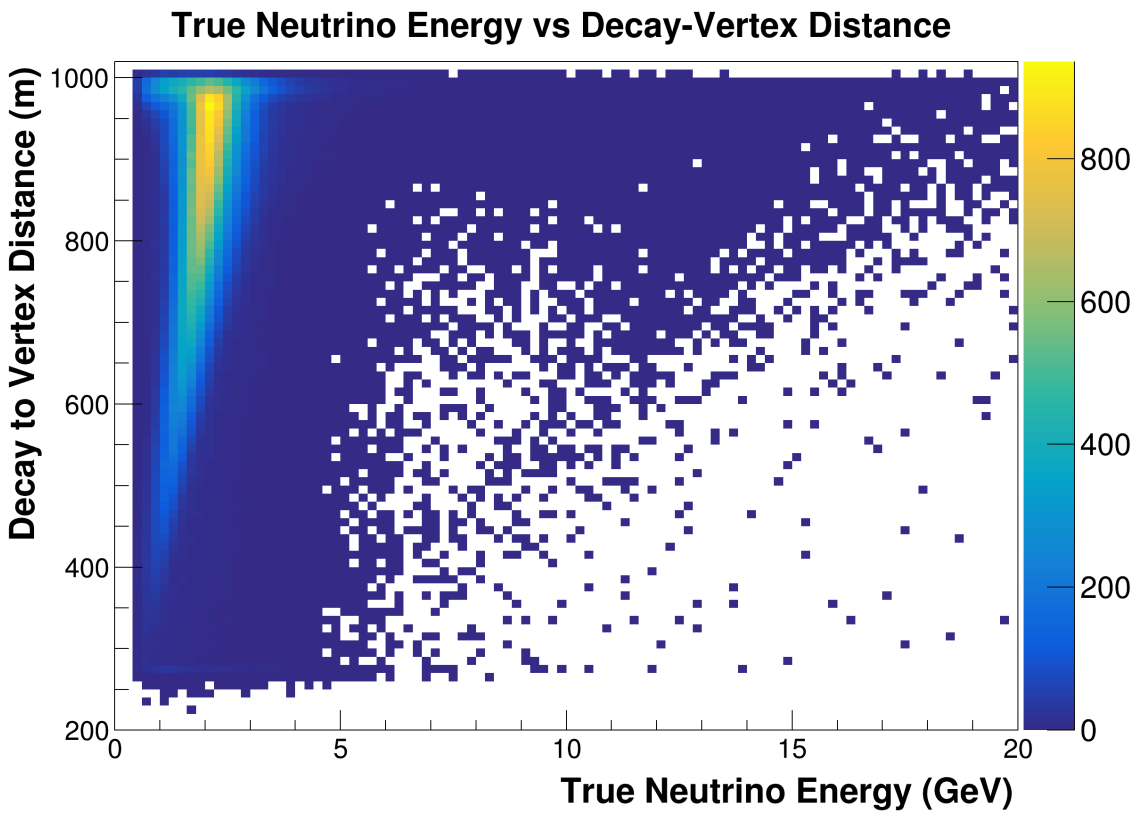
Sidereal vs. Solar Time





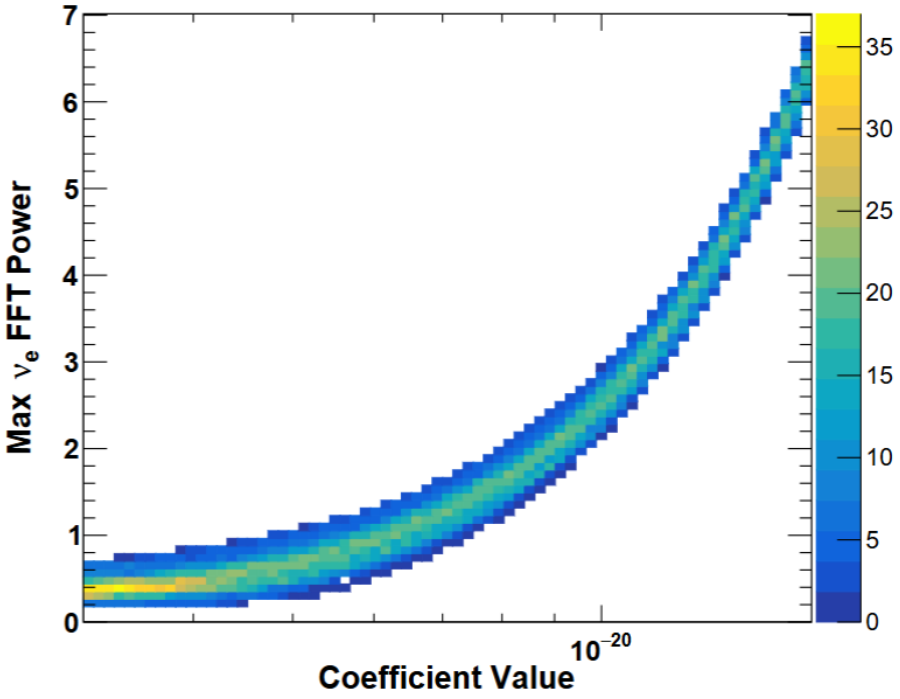
Backup: Energy-Baseline Distribution

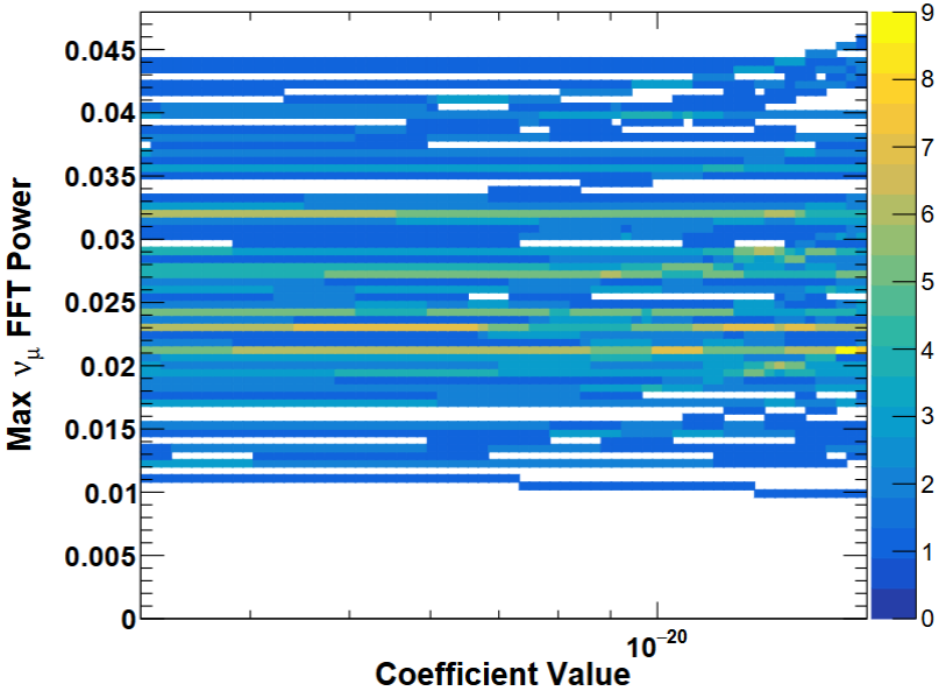
- Distribution of events vs true energy (GeV) and decay-vertex distance (m)
- Variation in L is significant at the ND due to short baseline
- Used to randomly sample E and L for LV-weighted simulated events



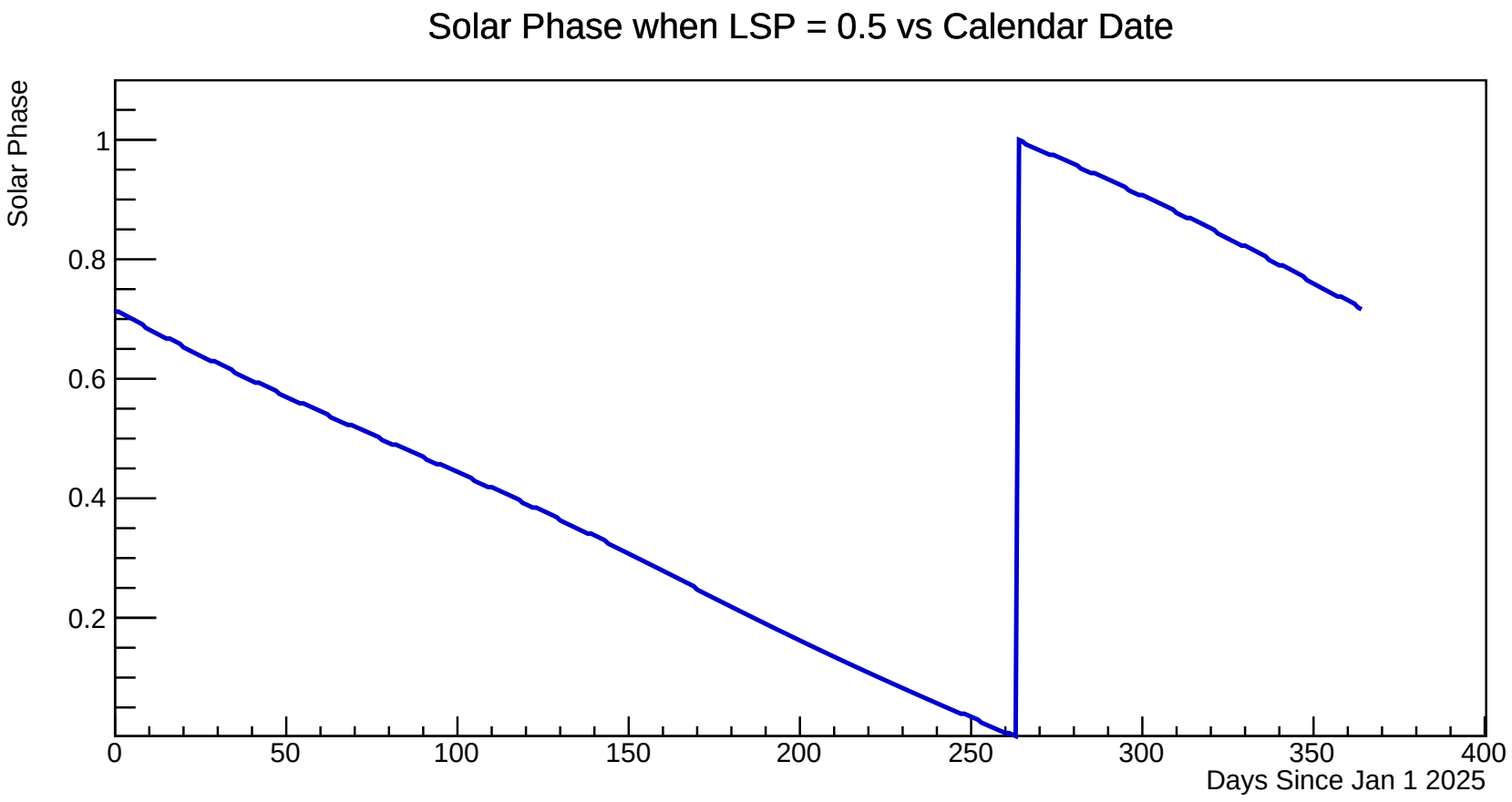
* *Histogram courtesy of Anna Cooleybeck, UW-Madison ν-group, 2025.*

Backup: FFT vs Coefficient Value



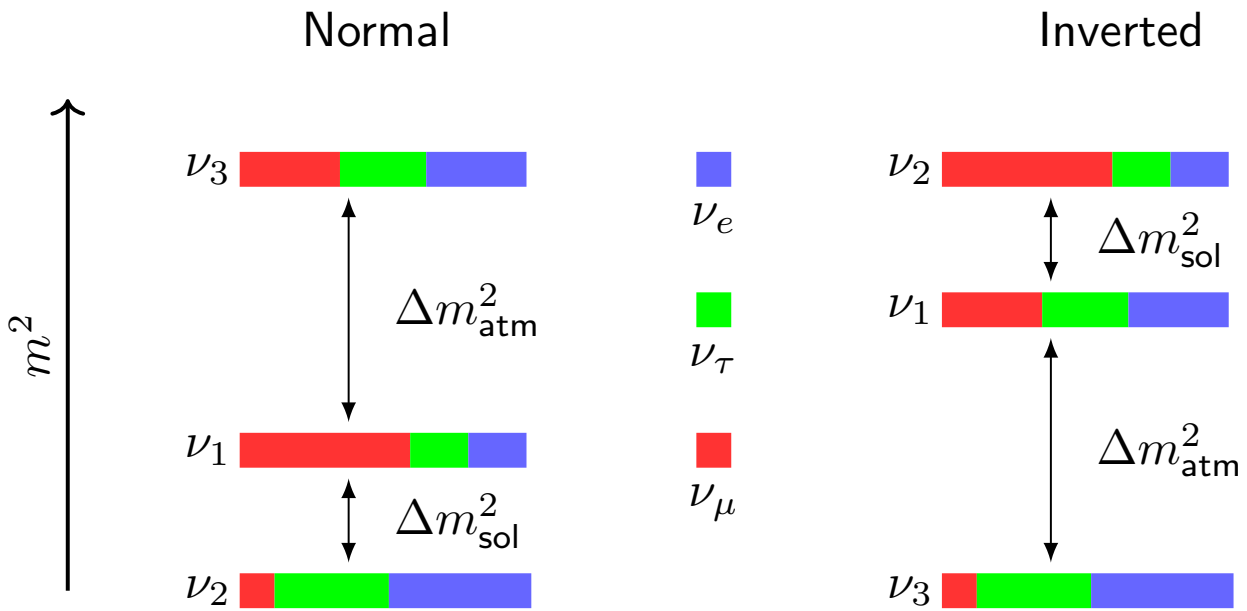


Backup: Sidereal vs. Solar Time Cont.



Backup: Neutrino Mass

- The Standard Model assumes massless neutrinos
- Solar neutrino studies prove that neutrinos have small, non-zero mass
- **Neutrino mass is direct evidence of physics beyond the Standard Model**



Backup: Neutrino Oscillation

- Neutrino flavor states are mixtures of mass states
- As they propagate, they 'oscillate' flavor states
- Observed in solar, reactor, astrophysical, and accelerator neutrinos
- **Neutrinos are sensitive to new physics**

